



Optimal lot reinspection based on truncated life test for logistic Rayleigh distribution

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Abstract:

Single sampling scheme is a conventional method for evaluating the equality of lots of products. Repetitive sampling plans are introduced in order to reduce the size of samples drawn from the lot. The product lifetime follows the logistic Rayleigh distribution. Best repetitive sampling plans are determined by minimizing the sampling effort using integer nonlinear programming. In terms of average sample number, optimal repetitive sampling plans outperform single sampling plans. A practical application is included for illustrative purposes.

Keywords: Quality control, Repetitive sampling plans, Average sample number, Producer and Consumer risks.

Mathematics Subject Classification (2010): 62N05, 62P30.

1 Introduction

Acceptance sampling plans are usually designed to decide the acceptability of submitted lots. In these plans, the lots of the product are tested using inspection schemes to determine if they are accepted or rejected based on random samples selected from them. Designing of inspection plans must provide appropriate protections to both producers and consumers while minimizing the

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sampling inspection effort to find the best plan. The constrained optimization problems can be solved to obtain the best inspection plan. Methods to determine sampling plans have been widely studied. Among others, we can cite [Naghizadeh Qomi and Fernández \(2024\)](#), [Naghizadeh Qomi et al. \(2025\)](#) and [Naghizadeh Qomi and Aslam \(2026\)](#).

Single sampling is the classical and most frequently used inspection scheme applied in practice. In this scheme, the lot is accepted if the number of failed products (defectives) in the selected sample is sufficiently low. However, in many cases, a single inspection can lead to occasional incorrect decisions about entire lots of products. In such situations, lot reinspection may be desirable to ensure the correctness of the decision. This type of plans are often more effective in reducing sample sizes. For example, repetitive sampling is an inspection method in which the acceptance and rejection of the lot is based on the repeated sample results of the same lot. The repetitive sampling plan (RSP) in lot inspection was introduced by [Sherman \(1965\)](#) and has been thoroughly examined by researchers. See [Jeyadurga and Balamurali \(2025\)](#), [Naghizadeh Qomi and Mahdizadeh \(2025\)](#) and [Naghizadeh Qomi et al. \(2026\)](#).

[Chaudhary and Kumar \(2020\)](#) introduced the logistic-Rayleigh (LR) distribution as a suitable model for real life applications and survival analysis. During the literature exploration, it was observed a lack of initiative in establishing RSPs for LR distribution. In this paper, we employ the repetitive methodology to design optimal time-censored (or type-I censored) reliability sampling plans with limited producer's and consumer's risks when the lifetime variable follows LR distribution.

The remainder of this work is structured as follows. A single sampling plan (SSP) based on LR distribution is presented in Section 2. In Section 3, we describe the RSP methodology for the assumed probability distribution along with their mathematical formulation. A descriptive and comparative analysis of the proposed inspection scheme is provided in Section 4. Section 5 uses a real data set to illustrate the proposed plan in real-life situations. Section 6 concludes the paper with a summary of findings and relevant considerations.

2 LR single sampling plans

Suppose that a reliability engineer wants to judge the acceptability of a large lot of products. Assume that the random variable T represents the product lifetime and also that the distribution of T is LR with the shape and scale parameters $\delta > 0$ and $\theta > 0$, respectively, i.e., $T \sim LR(\delta, \theta)$.

The probability density function (pdf) and the corresponding cumulative distribution function (cdf) of T are given respectively by

$$f(t; \delta, \theta) = \frac{\delta \theta t e^{\theta t^2/2} [e^{\theta t^2/2} - 1]^{\delta-1}}{(1 + [e^{\theta t^2/2} - 1]^{\delta})^2} \quad \text{and} \quad F(t; \delta, \theta) = 1 - \frac{1}{1 + [e^{\theta t^2/2} - 1]^{\delta}}, \quad t > 0. \quad (2.1)$$

The pdf of the LR distribution is right-skewed and then the median quality characteristic can be used to indicate the quality of product. The median of T is given by $\mathcal{M} = [(2 \log 2)/\theta]^{\frac{1}{2}}$.

Assume that p denotes the true proportion nonconforming. When the lot size is large, the number $\mathcal{D}$ of defective units follows approximately a binomial distribution, that is, $\mathcal{D} \sim \text{Bin}(n, p)$. An SSP would be characterized by a pair $\mathcal{P}_1 = (n, c)$ and the decision process associated with the inspection scheme $\mathcal{P}_1$ can be stated as follows: Step 1: Select n units at random from the submitted lot and count the number $\mathcal{D}$ of defective units. Step 2: Accept the lot if $\mathcal{D} \leq c$, where c is the so-called acceptance number and reject the lot otherwise. The operating characteristic (OC) function which is the lot acceptance probability is given by

$$\mathcal{L}_1(p) \equiv \mathcal{L}_1(p, n, c) = P(\mathcal{D} \leq c | p) = \sum_{i=0}^c \binom{n}{i} p^i (1-p)^{n-i}, \quad 0 < p < 1. \quad (2.2)$$

3 Optimal LR repetitive plans

In a standard RSP, the submitted lot is accepted if the number $\mathcal{D}$ of defective items in the chosen sample is small and rejected if it is large. Otherwise, the analyst has to repeat indefinitely the above process until a final decision is made. The procedure of a standard RSP can be described as follows: Step1: Select a sample of size n from the lot and put it on test for a pre-determined time t . Step 2: Accept the lot If $\mathcal{D} \leq c_1$ and reject if $\mathcal{D} > c_2$, where $c_2 \geq c_1$. Step 3: If $\mathcal{D} \in (c_1, c_2]$, then go to Step 1 and repeat the decision process with a new sample. The proposed RSP is characterized by $\mathcal{P}_2 = (c_1, c_2, n)$. If $c_1 = c_2$, this plan reduces to SSP.

The OC function associated with $\mathcal{P}_2 = (c_1, c_2, n)$ is defined as $\mathcal{L}_2(p) \equiv \mathcal{L}_2(p, n, c_1, c_2)$ which is given by

$$\mathcal{L}_2(p) = \mathcal{L}_1(p, n, c_1) \sum_{j=1}^{\infty} \{\mathcal{L}_1(p, n, c_2) - \mathcal{L}_1(p, n, c_1)\}^j = \frac{\mathcal{L}_1(p, n, c_1)}{\mathcal{L}_1(p, n, c_1) + 1 - \mathcal{L}_1(p, n, c_2)}. \quad (3.1)$$

Let $\mathcal{M}$ be the true median lifetime of products and $\mathcal{M}_0$ is the minimal median lifetime that is acceptable to the consumer and producer. The quality level of a product can be expressed as $r = \mathcal{M}/\mathcal{M}_0$. The termination time t can be determined as a multiple of $\mathcal{M}_0$, that is $t = d_0 \mathcal{M}_0$, where d_0 is the termination ratio. Moreover, the failure probability or proportion defective before time t is $p \equiv p(\delta, \theta, r)$ which can be obtained as $p = 1 - \frac{1}{1 + [\exp\{\frac{d_0^2 \log 2}{r^2}\} - 1]^{\delta}}$.

Assume that p_1 and p_2 are the rejectable and acceptable defective rates associated with the consumer and producer, respectively. The consumer and producer risks are defined as $\sup_{p \geq p_1} \{\mathcal{L}_2(p)\}$ and $\sup_{p \leq p_2} \{1 - \mathcal{L}_2(p)\}$, respectively. Since $\mathcal{L}(p)$ is a decreasing function of p , then the consumer and producer risks are given by $\mathcal{L}_2(p_1)$ and $1 - \mathcal{L}_2(p_2)$.

Let r_1 and r_2 be the ratio of quality level corresponding to the consumer and producer, respectively. The consumer demands that the lot acceptance probability should be smaller than the specified consumer's risk β at a lower quality level ($r_1 = 1$), whereas the producer requires that the lot rejection probability should be smaller than the specified producer's risk α at a higher quality level ($r_2 > 1$). The inspection cost of $\mathcal{P}_2 = (c_1, c_2, n)$ is measured by its average sample number (ASN). The ASN at $p \in (0, 1)$ is denoted by $ASN(p) \equiv ASN(p, n, c_1, c_2)$ and it is defined by

$$ASN(p) = n \sum_{j=0}^{\infty} \{\mathcal{L}_1(p, n, c_2) - \mathcal{L}_1(p, n, c_1)\}^j = \frac{n}{\mathcal{L}_1(p, n, c_1) + 1 - \mathcal{L}_1(p, n, c_2)} \quad (3.2)$$

Evidently, $ASN(p, n, c_1, c_1) \equiv n$, i.e., the RSP reduces to the SSP. We are interested to obtain the optimal repetitive plans $\mathcal{P}_2^* = (c_1^*, c_2^*, n^*)$ that satisfies the nonlinear inequality constraints $1 - \mathcal{L}_2(p_2) \leq \alpha$ and $\mathcal{L}_2(p_1) \leq \beta$. The constrained optimization problem is an integer nonlinear programming problem, which can be stated as follows:

$$\begin{aligned} & \text{Minimize} && ASN(p_1) \\ & \text{Subject to} && \mathcal{L}_2(p_1, n, c_1, c_2) \leq \beta, \\ & && \mathcal{L}_2(p_2, n, c_1, c_2) \geq 1 - \alpha, \\ & && 0 \leq c_1 \leq c_2 < n, \end{aligned} \quad (3.3)$$

where $n \in \mathbb{N} = \{1, 2, \dots\}$ and $c_1, c_2 \in \mathbb{N}_0 = \mathbb{N} \cup \{0\}$. A step-by-step iterative method to find the $ASN(p_1)$ -optimal RSP can be summarized as follows: Step 1: Set the value of parameter δ , producer's and consumer's risks (α, β) and the termination ratio d_0 . Step 2: Set the initial values of plan parameters $n = 1$, $c_1 = 0$ and $c_2 = c_1 + 1$. Step 3: Compute the probabilities $\mathcal{L}_2(p_1, n, c_1, c_2)$ and $\mathcal{L}_2(p_2, n, c_1, c_2)$ using equation (3.1) and $ASN(p_1)$ using equation (3.2). Then, check the inequalities $\mathcal{L}_2(p_1, n, c_1, c_2) \leq \beta$ and $\mathcal{L}_2(p_2, n, c_1, c_2) \geq 1 - \alpha$. Step 4: Obtain the different feasible plans (n, c_1, c_2) verifying the above inequalities. From the feasible solutions, find the best plans (n^*, c_1^*, c_2^*) that minimize the $ASN(p_1)$.

4 Descriptive and comparative analysis

The behavior of the proposed RSPs is analyzed in this section. The ASN-optimal repetitive LR plans (c_1^*, c_2^*, n^*) are presented in Tables 1 and 2 for $\alpha = 0.01$, $\beta = 0.05, 0.10$, $r_1 = 1$, $r_2 = 2, 3, 4, 5$,

Table 1: Optimal RSPs (c_1^*, c_2^*, n^*) for selected values of β, r_2 and d_0 when $\alpha = 0.01$ and $\delta = 1$.

β	r_2	$d_0 = 0.5$				$d_0 = 1$			
		c_1^*	c_2^*	n^*	ASN^*	c_1^*	c_2^*	n^*	ASN^*
0.05	2	2	5	41	59.8627	2	5	12	18.9849
	3	0	2	20	29.8870	1	3	9	11.7551
	4	0	2	20	29.8870	0	2	6	8.9302
	5	0	1	19	21.9298	0	1	5	5.9259
0.10	2	1	4	28	54.2724	2	5	12	18.9849
	3	0	2	17	29.5615	0	2	5	9.4118
	4	1	2	25	29.1073	1	2	7	8.3738
	5	0	1	15	19.0100	0	1	4	5.3333

Table 2: Optimal RSPs (c_1^*, c_2^*, n^*) for selected values of β, r_2 and d_0 when $\alpha = 0.01$ and $\delta = 2$.

β	r_2	$d_0 = 0.5$				$d_0 = 1$			
		c_1^*	c_2^*	n^*	ASN^*	c_1^*	c_2^*	n^*	ASN^*
0.05	2	0	2	97	140.8509	0	2	6	8.9302
	3	0	1	90	104.1581	0	1	5	5.9259
	4	0	1	90	104.1581	0	0	5	5.0000
	5	0	0	86	86.0000	0	0	5	5.0000
0.10	2	0	2	81	136.8967	0	1	4	5.3333
	3	0	1	72	90.5450	0	1	4	5.3333
	4	0	0	66	66.0000	0	0	4	4.0000
	5	0	0	66	66.0000	0	0	4	4.0000

$d_0 = 0.5, 1$ when $\delta = 1$ and $\delta = 2$, respectively. From Tables 1-2, it can be observed that when considering the same values of δ and r_2 , as the value of d_0 increases from 0.5 to 1.0, it becomes evident that there is a decrease in the magnitude of n^* and ASN^* . The optimal sample size n^* and value ASN^* generally becomes smaller when β increases. Moreover, ASN^* value tends to decrease as quality level r_2 increases.

We then analyzed how the proposed RSP outperforms the SSP based on LR distribution. The minimum ASN of the proposed RSP and the corresponding optimal sample size of SSP are plotted in Figure 1 for $d_0 = 0.5, 1.0$, $\alpha = 0.01$, $\beta = 0.05$ and $\delta = 1$. It is clearly evident that the proposed plan performs better than the SSP, in terms of ASN.

5 Real data application

The results obtained in this paper are illustrated with application on a real data set. The following observations studied by Nelson and Doganaksoy (2023) are from an accelerated life test of 59

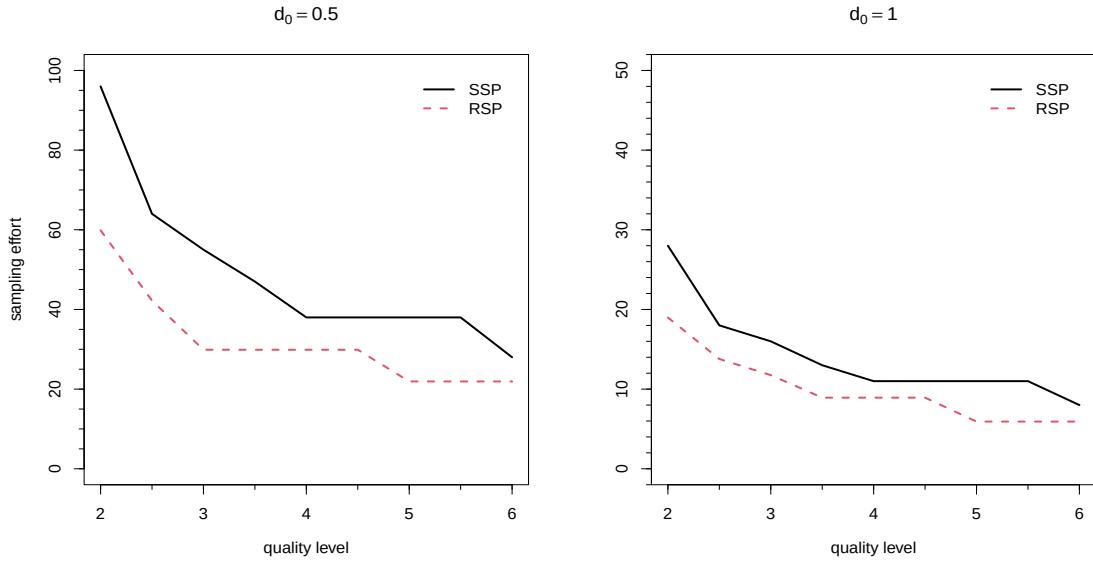


Figure 1: ASN curves for the SSP and the RSP versus the quality level r_2 when $\alpha = 0.01, \beta = 0.05, d_0 = 0.5, 1.0$ and $\delta = 1$.

conductors 2.997, 4.137, 4.288, 4.531, 4.700, 4.706, 5.009, 5.381, 5.434, 5.459, 5.589, 5.640, 5.807, 5.923, 6.033, 6.071, 6.087, 6.129, 6.352, 6.369, 6.476, 6.492, 6.515, 6.522, 6.538, 6.545, 6.573, 6.725, 6.869, 6.923, 6.948, 6.956, 6.958, 7.024, 7.224, 7.365, 7.398, 7.459, 7.489, 7.495, 7.496, 7.543, 7.683, 7.937, 7.945, 7.974, 8.120, 8.336, 8.532, 8.591, 8.687, 8.799, 9.218, 9.254, 9.289, 9.663, 10.092, 10.491, 11.038. We consider some well-known distributions in order to illustrate the goodness of fit of the LR as follows:

1. Logistic-Exponential (LE) distribution with pdf:

$$f_{LE}(t) = \theta \delta e^{\theta t} (e^{\theta t} - 1)^{\delta-1} \left\{ 1 + (e^{\theta t} - 1)^{\delta} \right\}^{-2}, \quad t > 0, \delta > 0, \theta > 0.$$

2. Generalized Exponential (GE) distribution with pdf:

$$f_{GE}(t) = \delta \theta e^{-\theta t} (1 - e^{-\theta t})^{\delta-1}, \quad t > 0, \delta > 0, \theta > 0.$$

3. Exponential power (EP) distribution with pdf:

$$f_{EP}(t) = \delta \theta^{\delta} t^{\delta-1} e^{-(\theta t)^{\delta}} \exp \left\{ 1 - e^{(\theta t)^{\delta}} \right\}, \quad t > 0, \delta > 0, \theta > 0.$$

The eligibility of data for the LR distribution was assessed using the Akaike information criterion (AIC), Bayesian information criterion (BIC), the log-likelihood (LL) function, Kolmogorov-Smirnov (K-S) statistic and p-value. Table 3 displays the model fitting summary of the chosen

Table 3: Distribution fit test results of considered data set.

Model	-LL	AIC	BIC	KS	p-value
LR	111.2049	226.4098	230.5649	0.0479	0.9983
LE	111.5138	227.0276	231.1827	0.0492	0.9975
GE	114.9473	233.8946	238.0497	0.1042	0.5103
EP	116.5015	237.0029	241.1580	0.1365	0.2021

data set. From Table 3, it is clear that the LR distribution is best fit for considered data set as compared to LE, GE and EP distributions. The maximum likelihood estimates of the parameters are $\hat{\delta} = 2.6968$ and $\hat{\theta} = 0.0290$. Using the fact $\mathcal{M} = [2 \log 2 / \theta]^{0.5}$, the median life can be estimated as $\mathcal{M}_0 = 6.914$. Considering the termination ratio for this data set as $d_0 = 1$, we get the termination time $t = 3.457$ minutes. Given $r_1 = 1, r_2 = 2$ and $(\alpha, \beta) = (0.01, 0.05)$. Then, the optimal RSPs are $(0, 1, 5)$ with corresponding ASN as 5.93. The decision process can be stated as follows:

- Step 1: Select a sample of size $n^* = 5$ from the submitted lot and put on test for time $t = 3.457$ minutes.
- Step 2: If there is no failure before time $t = 3.457$ minutes, accept the lot. If there is more than one failure, reject the lot and terminate the test.
- Step 3: If there is one failure, i.e., $\mathcal{D} = 1$, then go to step 1 and repeat the experiment.

Discussion and Conclusion

Time-censored repetitive sampling plans for LR distribution are introduced in this paper. The LR distribution is used to model the product time-to-failure. In our situation, the proposed plans depend on the median lifetimes specified by the producer and the consumer, for given maximum admissible sampling risks. The RSP that minimizes the ASN was derived by solving the corresponding integer nonlinear programming. From an inspection-effort perspective, the proposed RSPs outperform traditional SSPs. Traditional SSPs are simple but often require large sample sizes, resulting in high costs, especially for expensive or safety-critical products. RSPs typically reduce sample sizes.

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